Mian Saad Karim

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RESEARCH INTERESTS

Distributed Autonomous Robotic Systems, On-Device Edge AI / TinyML Perception, Multi-MCU Avionics Architectures, Non-Terrestrial Ad-Hoc Swarm Mesh Networks, Embedded Hardware-Firmware Co-Design, and Power Integrity for High-Noise Dynamic Platforms.

EDUCATION

Pakistan Institute of Engineering & Applied Sciences (PIEAS) Islamabad, Pakistan
Bachelor of Science in Electrical Engineering Graduated: July 2026

- **Cumulative GPA:** 3.64 / 4.00
- **Scholarship: USAID Merit & Needs-Based Scholar** (3-Year Award, ∼180k PKR/semester)
- **Undergraduate Thesis:** *UAV Swarm for Autonomous Disaster Assessment* *Supervisor: Dr. Waqas Ahmad*
- **Selected Coursework:** Embedded Systems Design, Digital Signal Processing, Linear Control Systems, Computer Architecture, Power Electronics, Digital Logic & FPGA (Verilog), Microwave & RF Engineering, Robotics & Autonomous Navigation.

RESEARCH MANUSCRIPTS & SUBMISSIONS

- **Mian Saad Karim**, Laraib Fatima, Muhammad Khuzaifa, Waqas Ahmad, *"A Dual-MCU Flight Controller Architecture for Onboard Edge AI and UAV Swarming Applications,"* **Submitted** to the *International Conference on Frontiers of Information Technology (FIT)*, COMSATS / IEEE, 2026.
- Laraib Fatima, **Mian Saad Karim**, Muhammad Khuzaifa, *"UAV Swarm for Autonomous Disaster Assessment,"* PIEAS Undergraduate Engineering Thesis, Supervised by Dr. Waqas Ahmad, June 2026.

FEATURED RESEARCH & TECHNICAL PROJECTS

UAV Swarm for Autonomous Disaster Assessment *PIEAS Capstone Thesis*
Sole Hardware/Firmware Architect in a 3-person team; 100% ownership of PCB design, firmware drivers, and bring-up [Code Repository]

- **Asymmetric Multi-MCU Architecture:** Designed a custom 4-layer FR4 flight computer partitioning mission-critical real-time flight stabilization (STM32F405, 168 MHz Cortex-M4F executing ArduPilot at 400 Hz) from non-deterministic Edge AI vision and swarm networking (ESP32-S3, 240 MHz dual-core Xtensa LX7 running custom ESP-IDF / FreeRTOS).
- **Onboard Neural Perception:** Deployed an Int8-quantized FOMO CNN model accelerated with ESP-NN SIMD vector instructions, processing 96×96 grayscale DVP camera feeds via DMA in 100 ms per inference window. Implemented temporal confidence hysteresis filtering to eliminate transient false positives in dynamic outdoor lighting.
- **Decentralized Non-Terrestrial Swarm Mesh:** Implemented a peer-to-peer ESP-NOW mesh layer allowing direct drone-to-drone survivor telemetry and GPS target sharing across 2 physical UAVs + 1 ground bridge without Wi-Fi APs or cellular infrastructure.
- **Power Integrity & Noise Suppression (Hard Problem Solved):** Identified high-current BLDC motor inductive switching spikes (up to 30 A transient steps) causing 3.3 V MCU rail brownouts and IMU yaw drift. Redesigned board with single-point star grounding, physical separation of noisy power traces from analog sensor planes, LC pi-filters, and fast TVS/Schottky flyback suppression.
- **Experimental Validation:** Validated through autonomous waypoint execution, survivor identification, and live telemetry streaming over WebSockets.

Smart Pen Micro-Motion Classification & Handwriting TinyML *Freelance Client: sbkaufman (5.0 ★)*

- **Time-Series Signal Processing:** Designed a high-rate data logging pipeline capturing 6-axis IMU (gyroscope + accelerometer) signals at 100 Hz across 10 writers tracing 20 distinct lexical categories.
- **Feature Engineering & TinyML:** Extracted spectral power, time-domain variance, and zero-crossing rates; trained a 1D CNN + GRU model achieving **95.9% classification accuracy** on held-out, unseen test writers.
- **Bare-Metal Deployment:** Quantized model to Int8 via TensorFlow Lite Micro and deployed on a resource-constrained microcontroller using double-buffered circular DMA buffers with sub-35 ms latency.

Modular 3-PCB Flight Stack (SpeedyBee F405 Wing Architecture Clone) *Freelance Client: awesumaiden (5.0 ★)*

- Designed an integrated 3-board modular stack: Flight Controller (STM32F405RGT6 + ICM-42688-P low-noise IMU + AT7456E SPI OSD), Power Distribution Board (2–6S LiPo input with 5V/9V synchronous buck regulators), and isolated USB-C breakout.
- Optimized differential signal routing and ground return paths to eliminate video noise and maintain sensor fidelity under high-load motor transients. Validated for INAV autonomous fixed-wing navigation.

FPGA Hardware Accelerators & Digital Logic Synthesis [GitHub: Servo Control] · [GitHub: VGA Arcade]

- **Dual-Axis Servo PWM Accelerator:** Authored synthesizable Verilog RTL on Xilinx Spartan-3E FPGA implementing precise sub-microsecond dual-axis PWM controllers, a hardware UART transceiver FSM, and a custom Python Tkinter GUI interface.
- **RTL 640×480 VGA Video Engine:** Implemented cycle-accurate horizontal/vertical sync generators and active frame memory logic purely in Verilog RTL to render an interactive multi-object flight arcade game without soft-core CPU overhead.

PROFESSIONAL & RESEARCH EXPERIENCE

QuickGen Technologies Islamabad, Pakistan
Embedded Systems Engineer Aug 2026 – Present

- Translating client product concepts into end-to-end hardware solutions; designing an AI-enabled calculator (STM32 + Raspberry Pi CM5 in calculator form factor) and a screenless NFC-driven children's smart speaker (ESP32-S3 + SD card audio).

Signatics Islamabad, Pakistan
Hardware Design & Verification Engineer July 2026 – Present

- Member of a 10-engineer Hardware Verification Department within a 60-person firm; responsible for characterizing high-speed embedded platforms, software-defined radio (SDR) motherboards, and commercial electronics.
- Architecting custom multi-rail DC-DC power delivery boards with multi-stage TVS overvoltage suppression and reverse-polarity protection.
- Developing automated test fixtures and test scripts to validate rail ripple, thermal derating, and signal integrity under extreme load cycles using high-bandwidth digital storage oscilloscopes (DSOs), logic analyzers, and electronic loads.

Saf Shikan Islamabad, Pakistan
Electrical Engineer (Part-Time: Swarm UAV Systems) Sept 2025 – June 2026 · *Intern* July 2025 – Sept 2025

- Spearheaded development of custom ESP32 mini-UAVs; designed schematic and PCB layout integrating 6-axis IMU, optical flow velocity sensors, and ultra-compact motor drive circuitry with Wi-Fi / ESP-NOW telemetry.

NASTP (Quest Aerospace Program) July 2025 – Aug 2025 · **PAC Kamra (Avionics)** July 2024 – Aug 2024

- Prototyped aerospace STEM educational kits (secured NASTP budget funding); gained avionics manufacturing, RF testing, and environmental stress screening experience at PAC.

HONORS, AWARDS & FELLOWSHIPS

- **USAID Merit and Needs-Based Scholarship:** Awarded by USAID and Higher Education Commission (HEC); highly competitive 3-year award (∼180k PKR/semester) covering full tuition, boarding, and monthly stipend throughout undergraduate studies (2023–2026).
- **Runner-up, Electro Masters Competition:** IEEE International Students, Young Professionals, and Women in Engineering Congress (ISYWSC'24) – Recognized for innovative hardware design and rapid embedded problem solving (Dec 2024).
- **4.8 / 5.0 Star Rated Embedded Engineering Consultant:** Fiverr Platform – Delivered verified solutions in TinyML gesture recognition, high-density flight controller PCBs, and custom FreeRTOS firmware for global clients.
- **Delegate, International Space & Youth Conference (ISYWC):** Selected participant representing PIEAS in aerospace robotics.

TECHNICAL COMPETENCIES

PCB Design & Power Integrity: 2-layer and 4-layer PCB design (Altium Designer, KiCad, EasyEDA); DFM/DFA, star grounding, LC pi-filters, TVS surge suppression, synchronous buck-boost (TPS63020), LDOs, inductive flyback clamp, rail ripple testing.

Embedded Systems & Firmware: STM32F405 (ARM Cortex-M4F), ESP32-S3 (Xtensa LX7), Bare-metal C/C++, ESP-IDF, FreeRTOS, ArduPilot (custom hwdef), INAV, DMA circular buffers, MAVLink v2, ESP-NOW ad-hoc mesh, SPI, I^2C, UART, RS-485, USB-C.

Edge AI & TinyML: Edge Impulse, TensorFlow Lite for Microcontrollers (TFLM), Int8 quantization, FOMO object detection, MobileNetV2, 1D CNN + GRU time-series classification, MFCC audio acoustic DSP feature extraction.

FPGA & Lab Instrumentation: Synthesizable Verilog HDL (Xilinx Spartan-3E, FSMs, PWM, VGA engine); Digital Storage Oscilloscopes (DSO), Logic Analyzers, Spectrum Analyzers, RF Test Stations, Variable DC Supplies, SMD Hot-air Rework.

LEADERSHIP & ACADEMIC SERVICE

- **Capstone Hardware Lead, PIEAS (2025–2026):** Mentored junior students in schematic capture, PCB layout rules, and debugging oscilloscope waveforms; managed project procurement and BOM logistics.
- **Peer Mentor in Embedded Systems:** Conducted peer study workshops on microcontrollers, FreeRTOS scheduling, and Altium Designer routing for electrical engineering undergraduates at PIEAS.